

Exercise 1: Fundamentals: SELECT & Filtering

SELECTING & Sorting

Retrieve all columns from the employees table.

① `SELECT * FROM employees`

id	first_name	last_name	department	salary	hire_date
1	Alice	Green	IT	70000	
2	Brian	Lee	HR	45000	
3	Cathy	Zulu	Finance	65000	
4	David	Makera	Marketing	50000	
5	Eva	Naidu	IT	72000	

② Find all unique departments

`SELECT DISTINCT department
FROM employees;`

department
IT
HR
Finance
Marketing

③ Retrieve the first and last names ordered by salary department

`SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;`

first_name	last_name	Salary
Eva	Naidu	72000
Alice	Green	70000
Cathy	Zulu	65000
David	Makera	50000

04 Retrieve the top 3 highest paid employees

```
SELECT l.d,  
       first_name, to  
       last_name,  
       salary  
FROM employees  
*WHERE salary > 6000*  
ORDER BY salary DESC  
LIMIT 3;
```

l.d	first_name	last_name	salary
5	Eva	Naidu	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

05 find employees in the IT department

```
SELECT l.d, first_name, last_name, department  
FROM employees  
WHERE department = 'IT';
```

l.d	first_name	last_name	department
5	EVA	NAIDU	IT
1	ALICE	GREEN	IT

06 Find employees in Finance with salary > 60000

```
SELECT l.d, first_name, last_name, department; salary  
FROM employees  
WHERE salary > 60000 AND department = 'finance';
```


id	first_name	last_name	department	salary
3	Cathy	Zuk	Finance	65000

07 Find employees in HR or Marketing

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('HR', 'Marketing');
```

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoen	Marketing

08 Find employees not in IT

id	first_name	last_name	department
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```
SELECT id, first_name, last_name, department
FROM employees
WHERE department <> 'IT'
```

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zuk	Finance
4	David	Mokoen	Marketing

09 Find employees in IT, HR, or Finance

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance')
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidu	IT

10 Find employees in IT with salary > 65000 and city Johannesburg

id	first_name	last_name	department	salary	City
1	Alice	Green	IT	72000	Johannesburg
5	Eva	Naidu	IT	72000	Johannesburg

```

SELECT id, first_name, last_name, department, salary, city
FROM employees
WHERE department = 'IT'
AND salary > 65000
AND city > 'Johannesburg';

```